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Course - Data Science

Assignment no: Research Assignment 2

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THE LANGUAGE OF BUSINESS ANALYTICS

### SECTION A

①. Business analytics - It refers to the use of data, quantitative analysis and stats methods to support decision making.

- ~~Data~~ Data being transformed into actionable insights.

Example: Quantitative analyst in asset management using stock prices to determine trends, performance, drawdowns and which economic conditions the stock perform best.

Reference: Sharda, R., Delen, D. (2020)

Business Intelligence, Analytics and Data Science: Managerial perspective: 4th Edition e.d.u.Peter

(2) Data Analytics - process of using raw data to uncover trends, patterns, market movement for decision making.

Example: Traders using Gold prices to determine trends, consolidations and reversals

Reference - Provost, F. and Fawcett T. (2013)

Data Science for Business, O'Reilly Media.

(3) Business Intelligence - The use of tools like Power BI to analyse data to support business decision making.

Example: Kofuwa using Power BI to determine how many students complain negatively or positively



(2)

Reference: Negash, S. (2004) Business Intelligence, Communications of the AIS, 13 (1) pp. 177-195.

(4) Data Driven Decision - refers to decisions taken after full data analysis was conducted  
Example: Multi-choice taking decisions on ads after Rofhwa conducted the analysis and has dashboards.

(5) Big Data - Refers to large complex datasets that has lots of rows and columns and requires advanced tools for processing.

Example: Meta process millions of users' interactions daily

Reference: Marr, B. (2016) Big Data Practice, Wiley.

(6) Descriptive analytics: the process of using historical data to study what happened to end up being at a certain point or achieving certain results.

Example: Khutsu using historical stock price data and macroeconomic factors to study how the increase of interest rates affect stock prices

Reference: - Sharda, R. et al (2020)

(7) Diagnostic Analytics - refers to the process of examining data to determine why something happened.

Example: Analysing why people dropped out of Bright Learn in data science class.

References - Alpaydin, E (2020) Introduction to Machine Learning. MIT Press.



(3)

(8). Predictive Analytics: the process of using ~~new~~ things like machine learning to predict future outcomes of certain things.

Example: ~~Good~~ Quantitative analyst using regression model or Lgboost model to predict price movements.

Reference: Siegel, E. (2026) Predictive Analytics. Wiley.

(9). Prescriptive Analytics: the process of recommending certain actions to achieve desired outcomes based on data analysis.

Example: Traders optimizing trades patterns and volumes to determine when to trade and reduce costs of trading.

Reference: Davenport, T. H. (2014) Big Data at Work. Harvard Business Review Press.

(10) Data Literacy - This is process of teaching, being able to read, analyze, understand and communicate data to people who can't even read it and who can read it effectively.

Example: Prothiwa in class teaching dashboards reports to everyone who has experience and doesn't have experience.

Reference: Wolff A. et al (2016) Creating an understanding of data literacy, Journal of Community Informatics, 12 (3).



## SECTION B

### DATA MEASURING AND REPORTING

11 Structured Data - Data that has order, ~~with~~ organized and stored in a formal format, typically in database.

Example: Matrix results stored according to schools, regions, provinces in databases.

Reference: Kitchin, R. (2014) The data Revolution, Sage.

(12) Unstructured data - this is data that has no order, or follow a certain formal pattern. the dataset is difficult, complex and hard to analyse.

Examples: Emails, videos, social media post and photos.

Reference: Nemakula, R. (2026). Databases Class. BrightLearn Portal.

(13) Data Warehouse - this is a central place where we store multiple databases using ETL and ELT process to clean, prepare, and organize data to be usable by data scientists or data analysts.

Example: McDonalds using different databases and store integrated data across the country in a data warehouse to analyse, ~~what gender~~ different datasets on segments, age and other factors.

Reference - Inmon, W. H. (2005) Building the Data Warehouse. Wiley.

(14) Data Mining: this is the process of identifying different patterns, their relationship with others in large datasets.



(5)  
Example: Checks identifying if women buy pads with tampons together most of the times or what.

(15) Data Visualization: this is where data analysts uses tools like power BI to display graphs, charts and trends to communicate insights.  
Example: Charts and graphs showing asset allocations, manager performances, stock performance and stock trends.

(16) Dashboards - these are the results of data visualizations that display key information and metrics monitoring performance of companies.  
Examples: Company dashboard displaying sales, growth, expense reductions, stock price growth, and profit growth.

(17) Metric :- these are numerical measures quantifiable, used by companies to track performance, sales growth, progress of the company, and how healthy are financial statements.

Example: Financial Metrics like, Net profit Margins, operating cashflow, Gross Profit Margin, EBITDA growth, Return on Investments and Return on Equity.

Reference: Parmenter, D. (2015) Key Performance Indicators. Wiley.



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(18). Key Performance Indicator (KPI) - this is a critical company metric normally used when companies give bonuses and annual increases.  
- Companies use this metric to measure the strength and weakness of the company against its peers.  
Example - Old Mutual using KPI to assess annual performance bonuses for employees.

19. Benchmark - this is a standard reference point that the companies or portfolio managers use to compare against performance.  
- In asset management, we use benchmarks to measure if the managed fund can beat its benchmark.

Example - BlackRock using MSCI WORLD Index to measure if their enhanced strategies can beat the benchmark.

- ~~Camp~~ Clicks comparing itself performance with other retailers (like Dischem) average.

Reference - Camp, R.C. (1999) Benchmarking.  
ASQC Quality Press

(20) Segmentation - this is process where market is divided in small segments or characteristics of their own group. Target market is divided according to character.

Example - Target market divided according to their income, gender or age.

Reference: Kotler P. and Keller K.L. (2016) Marketing Management. Pearson.



## SECTION C - CUSTOMER AND SALES KPI

(7)

(21) Conversion Rate - this is a percentage of visitors ~~out of~~ who take action against the number of visitors to a certain store.

- In Soccer is number of goals scored against the number of shots by players.

Example - If Barcelona total shots is 10 in a game and they score 3 goals, then the conversion rate is 30%.

Reference - Chaffey, P. et al (2019) Digital Marketing 7th edn. Pearson.

(22) Customer Acquisition Cost - this is a total cost of acquiring a customer including all the costs. The costs includes marketing, branding, sales expenses and other operating expenses.

Example - If Pick n' pay spend R20 000 trying to get customers and 200 customers the  $CAC = R100$

~~Reference~~ Reference: Kotler, P. and Keller K.L. (2016) Marketing Management. 15th edn - Pearson.

23 Customer Lifetime Value - this is the total revenue the company expects from its relationship with different customers.

Example - If Khutso spends R2500 on groceries every month at Checkers, then his yearly CLV is  $R2500 \times 12 = R30\ 000$  depends on number of years he buys at Checkers until he moves on to Woolworths.

Reference - Kumar, V. (2018) Customer Relationship Management. Springer.



(24) Churn rate - this is percentage of customers who move on from a product or service for different reasons. Could be treatment or product is obsolete.

Example - When people move on from Brightlearn because Rofliwa is giving them lots of assignments to Coursera. If 340 people leave Brightlearn out of 350 students - Churn rate =  $340/350 = 97.14\%$

Reference - Farris, P.W et.al. (2010) Marketing Metrics. Pearson.

(25) Customer Retention Rate - this is the percentage of students or customers stay for different reasons.

Example - Students who remain with Brightlearn if 10 students remain out of 350 then  $CRR = 10/350 = 2.8\%$

(26) Average Order Value - this refers to the average amount spent per transaction.

Example - If sales amount to R10000 and there has been 100 transactions, the  $AOV = R10000/100 = R100$ .

Reference - Chaffey, D. (2015) Digital Business and E-Commerce Management. Pearson

(27) Net Promoter Score - kind of month metric that measures customer loyalty and if customers will recommend the product or service.

Example - Likelihood of students recommending Brightlearn to their peers.



(28) Sales Funnel - This is the process that the customers go through from the day they are aware of the product until they buy a product.  
Example - Khutso became aware of Brigatlearn in June 2025 but registered in April 2026.  
Reference - Kotler, P. et al. (2017) Marketing 400 Wiley.

### SECTION D

(29) Marketing Campaign - marketing activities that are aimed at making customers aware of a certain product or service.

Example - 27 Four Investment Managers holding a campaign to promote Beta Solutions Strategies

(30) Post Campaign Analysis - Evaluating how marketing campaign performed and if benchmark was outperformed or underperformed.

Example - Measuring asset under management of Beta Solutions Strategies growth after the Campaign.

(31) Incremental Revenue - this is the revenue generated from the Campaign or investor day done by a business.

Example - Measuring revenue made after Aum was received by 27 Four ~~from~~ after Campaign held.

If the business generated R20m then now its generating R25m. then incremental revenue is  $R25m - R20m = R5m$ .

Reference - Faris, P.W et al. (2010) Marketing Metrics. Pearson.



(32). Return on Investment - it is a financial metric used to calculate how much was generated from the investment that was made.

Example - If Khutso invested R1000 into property investment and at the ~~end of~~ investment earns R500 to total R1500 then.

$$ROI = \frac{500}{1000} \times 100 = 50\%$$

(33). Return on Ad Spend - this is the revenue generated from advertising only.

Example - Spend R1000 on advertisement then generate R2000 then ROAS = 2:1

Reference - (Chaffey, D. (2019) Digital Marketing. Pearson

(34). Click-Through Rate (CTR) - this is the number of people who click on an advert after liking it over the number of people who saw the advert. Some don't click the ad.

Example - If 1 person clicks on Woolworths Ad out of 5 people who saw it. Then CTR =  $1/5 = 20\%$ .

Reference - Ryan, D. (2016) Understanding Digital Marketing. Kogan Page.

(35). Bounce Rate - This is the number of people who leave a website after view once of one page and not be interested on others over people who view website pages.

Example - If 20 visitors leave immediately after visiting online shopping website out of 200 visitors the Bounce Rate =  $20/200 = 10\%$



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(36) A/B Testing - this is the method of comparing two webpages and decide which one performs better or testing if course on a certain platform is better than the other.  
Example - testing if udemy is better than BrightLearn.

Reference - Kohavi, R et al. (2020) Trusting Online Controlled Experiments. Cambridge University Press.

(37) Lead Generation - this is the process of attracting customers ~~into~~ and capturing potential customers' interest for future buy of product or service.

Example - Cologne testing at malls to create awareness and attraction of customers' interest for future buy or

(38) Marketing Attribution - process of determining which form of marketing gets credit due to a certain buy after customers sees an Ad.  
Example - Woolworths crediting Ticktok for certain purchases in the business.

Reference - Li, H. and Kannan, P.K (2014) Attribution modeling, Journal of Marketing Research, 51 (1).